Whole-Body Retargeting for MABEL:

Clutch-Relative and Shoulder-Anchored Wrist Mapping, Cosine-Directional Arm IK, the Decoupled Look-At Neck, and the Lazy Reach Followers

(`mabel.whole_body_retargeting`)

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Abstract

This subpackage maps the operator's body pose — head and wrist transforms plus optional named hand skeletons, from *any* source (Vision Pro ARKit or a webcam skeleton via the web studios; `mabel.sources` normalises both) — onto MABEL's body targets. Every piece is decoupled and independently usable: three wrist-pose retargeting laws (a clutch-relative UMI-style mapping with per-operator scale, a per-shoulder extension-fraction mapping, and — the live default since 2026-07-24 — a *root-uniform scaled-keypoint* mapping in the style of *Retargeting Matters*/GMR [4], which preserves the operator's head–wrist–wrist pose *triangle* so the robot tracks the human's *scale*), a cosine-directional contact-point arm IK for parity with the apps' engine, a 3-DOF gaze look-at neck controller that reads only the head pose and writes only the neck (verified), and the lazy base/lift/torso followers that recruit the big DOFs only when the hand centroid leaves a comfort box. Per-operator anthropometry (the scale γ, shoulder anchors, finger sizes) comes from the calibration pipeline documented separately in `calibration.pdf`. Finger retargeting is deliberately *not* here (hands are position-controlled and decoupled, `hand_retargeting.pdf`). We also report and fix a pervasive redundant-arm *limit cycle* (a trust-region artifact) that made held poses visibly oscillate — the reported instability.

Keywords: teleoperation retargeting, general motion retargeting, scaled keypoint mapping, workspace mapping, look-at control, mobile manipulation, embodiment transfer, redundancy resolution.

1 Structure

module	class	role
`teleop_engine/`	`TeleopEngine`	GMR scaled-keypoint wrist map (live; §4)
`wrist_clutch.py`	`Retargeter`	wrist SE(3) targets (clutch / shoulder-anchored)
`arm_cosine.py`	`ArmRetargeter`	directional contact-point arm IK (damped GN)
`neck.py`	`NeckController`	decoupled 3-DOF gaze look-at
`followers.py`	`LazyBase/Lift/Torso`	comfort-box reach assists
`calibration.py`	`OperatorProfile`	per-operator anthropometry (see `calibration.pdf`)

Two arm paths coexist deliberately (flagged in `mabel_controller.pdf`): the *DLS path* — `Retargeter` produces palm SE(3) targets consumed by the differential IK (`whole_body_controller.pdf` §2) — and the *cosine path* — `ArmRetargeter` solves arm joints directly, matching the apps' engine. The neck and the followers compose with either.

2 Wrist mapping (`wrist_clutch.py`)

Clutch-relative position (default). At (re-)calibration the operator wrist w_0 and robot palm r_0 are anchored; thereafter

$$p^{\text{ee}} \;=\; r_0 \;+\; S \odot M\,(w - w_0), \tag{1}$$

where M is the fixed ARKit→robot axis remap and S the per-axis workspace scale. The deliberate amplification ($S = (1.3, 1.3, 1.4)$) keeps the operator's hands in a small comfortable box while the lazy followers (§7) carry large reaches. With a calibrated profile the scale is corrected by arm length relative to the reference operator, $S \leftarrow S \cdot \text{clip}(L_{\text{ref}}/L_{\text{user}})$, so the hand-tuned feel is preserved for the reference operator and adjusted for others (`calibration.pdf`). In *body-relative* mode the anchor r_0 is stored in the chassis frame and re-expressed through the current chassis pose each frame, so targets ride with the driving/turning base.

Shoulder-anchored position (option). Anatomically faithful reach:

$$p^{\text{ee}} \;=\; c_{\text{sh}}^{\text{rob}} \;+\; \gamma\, M\,\big(w - c_{\text{sh}}^{\text{hum}}\big), \qquad \gamma = \frac{\text{robot arm reach (live FK)}}{\text{operator arm length}}, \tag{2}$$

with the operator shoulder $c_{\text{sh}}^{\text{hum}}$ located from the calibrated head-relative offsets; falls back to Eq. (1) when inputs are missing.

Orientation. Relative-rotation clutch by conjugation:

$$R^{\text{ee}} \;=\; \big(M R_w R_{w_0}^{\top} M^{\top}\big)\, R_0, \tag{3}$$

(the rotation *angle* is conjugation-invariant, which the diagnostic HUD exploits for a 1:1 tracking check). In skeleton mode with a calibrated palm constant K, the absolute geometric match $R^{\text{ee}} = F_{\text{op}} K^{\top}$ is used instead, making the robot's natural palm frame equal the operator's. Grasp scalars derive from the calibrated pinch span.

3 Cosine-directional arm IK (`arm_cosine.py`)

A faithful port of the research stack's arm solver so the apps and the viewer produce matching motion. On a scratch `MjData` the arm joints q (plus the ORCA wrist as a jointly-optimised 8th DOF) minimise

$$\begin{aligned} E(q) \;=\; & W_{\text{wrist}}\,\big\|c^{\text{rob}}(q) - c^{\text{hum}}\big\|^2 \;+\; W_{\text{elbow}}\,\Big(1 - \hat{e}^{\text{rob}}(q)\cdot \hat{e}^{\text{hum}}\Big) \\ & +\; W_{\text{orient}}\Big[\big(1 - \hat{f}^{\text{rob}}\cdot \hat{f}^{\text{hum}}\big) + \big(1 - \hat{n}^{\text{rob}}\cdot \hat{n}^{\text{hum}}\big)\Big] \\ & +\; W_{\text{smooth}}\|q - q_{\text{prev}}\|^2 \;+\; \|q - q_{\text{nom}}\|^2_{W_{\text{reg}}}, \end{aligned} \tag{4}$$

solved under joint bounds — since 2026-07-22 by a projected damped Gauss–Newton with analytic Jacobians (§5, *Solver performance*; the damping keeps steps bounded at singularities), previously by SLSQP (35 iterations, warm-started; retained as the parity reference). The terms: the hand *contact point* $c = p_{\text{wrist}} + \ell\,\hat{f}$ ($\ell = 14\,\text{cm}$ along the palm forward axis, ≈ middle fingertip) is tracked in *position* (W=600, lets the two hands meet precisely); the elbow enters only as a *direction* from the shoulder and auto-disables when the source sends no elbow (ARKit); palm orientation enters as cosines of the forward and normal axes of the anatomically-constructed natural frame (identical

construction on operator skeleton and robot FK, `hand_retargeting.pdf` §2); and the per-joint posture regularisation is deliberately *asymmetric* ($W_{\text{reg}} = (0.08, 0.08, 0.40, 0.08, 0.02, 0.08, 0.60)$): hard on shoulder-roll and wrist-yaw so roll resolves at the forearm and wrist pitch — the way a human arm does — instead of spinning the shoulder.

4 GMR scaled-keypoint mapping (the live engine map)

The live engine (`mabel/teleop_engine/engine.py` — the controller the `MabelTeleop` facade composes for BOTH the simulated and the real robot) feeds the cosine arm IK of §3 wrist targets built by a *root-uniform scaled-keypoint* map in the style of *Retargeting Matters*/GMR [4]. It supersedes (2026-07-24) the per-shoulder extension map of §5, which superseded the legacy head-anchored map; all three are selectable (`MABEL_ARM_MAP`).

Why scale, and why a single root. The live requirement is that the robot reproduce the human's *pose shape*: the operator's three tracked upper-body coordinates — head and the two wrists — must sit, relative to one another, where the robot's do (up to the size difference). GMR [4] formalises this as a similarity transform about a common root. For a body b with per-body scale s_b and general size ratio h/h_{ref}, GMR places $p_b^{\text{tgt}} = (h/h_{\text{ref}})\, s_b\, (p_b^{\text{src}} - p_{\text{root}}^{\text{src}}) + p_{\text{root}}^{\text{tgt}}$ and *uniformly* scales the root translation — crucial to avoid foot sliding in their locomotion setting. MABEL has no legs to slide, but the same one-root similarity is exactly what preserves the head–wrist–wrist triangle. We take the root as the operator's **mid-shoulder** point $r_h = \frac{1}{2}(s_h^r + s_h^l)$ (`ShoulderEstimator.root_world`) and the robot root as its mid arm-mount r_r, one uniform upper-body scale $\sigma = L_r/L_h$ (robot usable reach over the operator's measured arm length, §5 Step 3; a single ratio, since MABEL's two arms are symmetric), and the body-frame rotation M_t of (12):

$$c_0 \;=\; r_r \;+\; \sigma\, M_t\, (w - r_h), \qquad \sigma = L_r/L_h. \tag{5}$$

Because a *single* (σ, M_t, r) acts on both wrists, the operator's inter-wrist vector maps to $\sigma M_t(w^r - w^l)$ exactly: the pose triangle is preserved by construction, at every separation and direction. This is the structural cure for the per-shoulder map's collapse (§5): anchoring each wrist on the robot's own shoulder — which is 3× narrower than a human's — distorts the triangle and drove clearly-separated human wrists to overlap on the robot, a defect that map could only paper over with an anti-collapse floor (Step 7). Measured (§5.2): the scaled-keypoint position error (the GMR $E_{\text{g-mpbpe}}$ analogue over head + two wrists) is 26 mm for this map versus 182 mm for the per-shoulder map and 92 mm for the head map, and the inter-wrist scale ratio is 0.965 (ideal 1.0) versus 0.541 and 0.902.

Extension renormalisation near full reach. A pure similarity does not by itself guarantee that a fully extended human arm fully extends the robot: L_h is the operator's own arm length, so $\sigma\|w - r_h\|$ at $\rho = \|w - s_h\|/L_h = 1$ need not sit at the robot's reach boundary once the root offset r is added. So above a gate $\rho \geq 0.85$ the target *magnitude about the robot's own shoulder* s_r is smoothstep-blended to the extension-exact radius $\rho L_r\, \text{od}(\rho)$ (the Step-6 law of the extension map, overdrive od and all):

$$\lambda = S\Big(\tfrac{\rho - 0.85}{0.15}\Big), \quad c = s_r + \hat{u}\,\big[(1-\lambda)\,\|c_0 - s_r\| + \lambda\,\rho L_r\,\text{od}(\rho)\big], \quad \hat{u} = \tfrac{c_0 - s_r}{\|c_0 - s_r\|}, \tag{6}$$

so mid-workspace is the pure pose-preserving GMR map and full extension keeps the extension map's one guarantee (measured $\rho_r = 0.96$ at full human reach, the elbow at its hardware stop).

The bimanual proximity blend and anti-collapse floor of Step 7 still apply, but are now essentially inactive (the triangle is already correct); they matter only at true hand contact, where (14) fades to the operator's 1:1 separation so touching hands still touch. Steps 1–5 and 8–11 below are shared verbatim (the body frame, shoulder anchors, arm length, synthesised elbow, palm orientation, the solver, and basin escape); only the wrist-target law changes.

5 Extension-fraction shoulder-anchored mapping (the prior map)

From 2026-07-21 to -24 this per-shoulder map was the live map, and it remains selectable (`MABEL_ARM_MAP=extension`); it is retained here both because it is still shipped and because Steps 1–11 document the shared pipeline the GMR map (§4) builds on. It feeds the cosine arm IK of §3 *extension-normalised* wrist targets instead of the legacy head-anchored map. Motivation, from a live operator report: with the head map, a fully extended human arm left the robot's elbow visibly bent. The head map places the target at $c = p^{\text{rob}}_{\text{head}} + \gamma_{xy}\, M_0\,(w - p^{\text{hum}}_{\text{head}})$ with the *horizontal* offset scaled by the calibrated reach ratio and the vertical mapped 1:1; because the robot's shoulder does not sit where the scaled human shoulder does relative to the head, the fully extended human wrist lands strictly *inside* the robot's reachable envelope and the IK tracks that interior point exactly — elbow bent. Measured (§5.2): interior elbow angle 121° at full human extension, 61° from the joint limit.

5.1 The mapping pipeline, step by step

Inputs per frame: the ARKit head pose $H = (R_H, p_H) \in SE(3)$ (world $+\hat{y}$ up), the per-hand named skeletons with wrist positions w^r, w^l, and the robot's live FK (chassis rotation R_{ch}, arm-mount positions $s_r^{r/l}$). All operator quantities are in the ARKit world; all robot targets are emitted in the robot world. Superscripts r/l are dropped where a step applies per side.

Step 1 — operator body frame B_{op}. A yaw-only, gravity-aligned frame with columns $[\hat{r}\ \hat{u}\ \hat{f}]$ (right, up, forward):

$$\hat{u} = \hat{e}_y, \qquad \hat{r} = \frac{A^{\top}\big[\sin\psi,\ -\cos\psi,\ 0\big]^{\top}}{\big\|\cdot\big\|}, \qquad \hat{f} = \hat{u} \times \hat{r}, \tag{7}$$

where ψ is the torso heading from the analytic body-frame estimator (head + two wrists; it rejects look-around, reach, and gaze-fixed motions that fool a head-only heading, and its wrist-consensus grounding makes it non-accumulating — `body_frame.pdf`) expressed through the fixed axis permutation A (`AVP_TO_ROBOT`); before the estimator is ready, $\hat{f}$ falls back to the head heading flattened to the horizontal, $\hat{f} \propto (I - \hat{u}\hat{u}^{\top})(-R_H \hat{e}_z)$. Head pitch/roll never tilt this frame, so a glance neither orbits the estimated shoulders (Step 2) nor rotates the map (Step 4).

Step 2 — stature and shoulder anchors. Stature from the HMD eye height with the Winter eye-height ratio, kept as a *running maximum* so crouching never shrinks the skeleton:

$$H_{\text{st}} \leftarrow \max\Big(H_{\text{st}},\ \text{clip}\big(p_{H,y}/0.936,\ 1.40,\ 2.10\big)\Big). \tag{8}$$

The shoulder centres are fixed offsets in the body frame (measured `OperatorProfile` values when present, Winter ratios otherwise: half-width $b = \frac{1}{2}\,0.259\,H_{\text{st}}$, eye→acromion drop $d = 0.118\,H_{\text{st}}$):

$$s_h^{r/l} \;=\; p_H \;\pm\; b\,\hat{r} \;-\; d\,\hat{u}. \tag{9}$$

Step 3 — operator arm length L_h. The calibrated profile length when measured; otherwise a live estimate floored by the anthropometric prior,

$$L_h \;=\; \max\Big(0.332\,H_{\mathrm{st}},\; \operatorname*{runmax}_{t}\,\mathrm{med}_5\,\|w - s_h\|\Big) \;\in\; [0.30, 0.90]\,\mathrm{m}, \tag{10}$$

where med_5 is the median of the last five samples (rejects single-frame ARKit spikes, which a raw running max would latch forever) and samples are discarded outside a plausibility band. The prior FLOOR matters: early sub-maximal reaches must not shrink L_h and over-read the extension; a longer-than-prior arm is adopted after its first full extension (measured: ρ_r at ρ_h=0.8 recovers from 0.95 to 0.82 after one full reach). A *shorter*-than-prior arm requires the measured profile (limitation, §Limitations). The 1-pose arms-forward calibration is MIRRORED (2026-07-23): the retarget runs alongside the capture, so the robot extends with the operator ($\rho_r = 0.96$ during the hold, its full-extension maximum) and L_h self-calibrates from those same frames — the earlier hold-still-while-measuring behaviour gave no visual confirmation and starved (10) of exactly the full-reach evidence it needs.

Step 4 — extension coordinates (the body-relative ground truth). Per side,

$$v = w - s_h, \qquad \rho = \min\Big(\frac{\|v\|}{L_h},\, 1\Big), \qquad \hat{n} = \frac{v}{\|v\|}, \qquad \chi = B_{\mathrm{op}}^{\top}\hat{n}, \tag{11}$$

i.e. the wrist is reduced to *how far along the operator's own arm it is* (ρ) and *which body-relative direction it points* ($\chi = [\chi_r, \chi_u, \chi_f]^{\top}$, components along the operator's right/up/forward). (ρ, χ) is invariant to where the operator stands and which way they face — this is the representation in which the mapping is defined, so it cannot "mess up over time": no world anchor is ever consulted.

Step 5 — the frame map M_t (operator body → robot body). The robot's semantic basis is built from its LIVE chassis rotation (R_{align} turns chassis axes into robot-body axes, x=fwd/y=left/z=up; the front is chassis $-x$):

$$W = R_{\mathrm{ch}}R_{\mathrm{align}}, \qquad R_{\mathrm{rob}} = \big[\, -W\hat{e}_2 \;\; W\hat{e}_3 \;\; W\hat{e}_1 \,\big] \;\text{(right, up, fwd)}, \qquad M_t = R_{\mathrm{rob}}\,B_{\mathrm{op}}^{\top}. \tag{12}$$

M_t maps any operator-world vector to the robot-world vector with the SAME body-semantic components. Invariance is immediate: a whole-body operator turn R_y sends $B_{\mathrm{op}} \to R_y B_{\mathrm{op}}$ and every skeleton vector $v \to R_y v$, so $M_t v \to R_{\mathrm{rob}} B_{\mathrm{op}}^{\top} R_y^{\top} R_y v = M_t v$; a chassis turn cancels identically on the robot side. Before the estimator has axes (or with `MABEL_ARM_BODY_FRAME=0`) M_t degrades to the fixed world permutation A — the pre-2026-07-22 behaviour, which E6 measures failing at π rad for a 180° operator turn.

Step 6 — raw wrist target. With L_r the robot's usable per-side reach (measured once by DLS from the arm mount: 0.669 m right, 0.660 m left),

$$c_0 \;=\; s_r \;+\; \mathrm{od}(\rho)\,\rho\,L_r\;M_t\,\hat{n}, \qquad \mathrm{od}(\rho) = 1 + 0.04\,S\Big(\frac{\rho - 0.85}{0.15}\Big), \tag{13}$$

with S the smoothstep $S(x) = \mathrm{clip}(x,0,1)^2\,(3 - 2\,\mathrm{clip}(x,0,1))$. The robot's wrist is commanded to the *same fraction of its own arm length, along the same body-relative direction from its own shoulder*: full human extension is full robot extension by construction, for every operator size and direction, and the target rides the robot's own body (lift/torso motion carry it). The overdrive od pushes the target marginally past the reach boundary *only* near $\rho = 1$, so the IK's position term overrules the posture regularisers and the elbow saturates its hardware range; mid-workspace stays exactly 1:1.

Step 7 — bimanual coupling (blend + anti-collapse floor). Per-shoulder anchoring is exact per arm but couples the PAIR only through anatomy, and the robot's biacromial width (0.148 m) is 3× smaller than a human's (∼0.45 m): inward-angled directions cross the robot's midline far sooner (measured 1.6 cm robot separation at 30 cm human — the reported wrist overlap). With $d = w^r - w^l$ (operator, metric):

$$\omega = S\Big(\frac{0.35 - \|d\|}{0.35 - 0.12}\Big), \qquad c^{r/l} \leftarrow (1-\omega)\, c^{r/l} \;+\; \omega\Big(\bar{c} \pm \tfrac{1}{2} M_t\, d\Big), \tag{14}$$

$$F = \min\big(0.18,\ \|d\|\big), \qquad c^{r/l} \leftarrow \bar{c} \pm \tfrac{1}{2} F \hat{a} \quad \text{iff} \quad \|c^r - c^l\| < F, \tag{15}$$

where $\bar{c}$ is the pair midpoint and $\hat{a}$ the current separation axis. (14) fades the pair to the operator's EXACT 1:1 metric separation near contact (touching hands touch); (15) is the anti-collapse floor — deliberately *absolute*, not γ-proportional: it can never push a target whose natural separation already exceeds a hand clearance, so one-arm tasks with the other arm hanging are untouched (a $\gamma\|d\|$ floor measurably contaminated them: full-reach ρ_r rose 0.959 → 0.988 and circle tracking tripled its error), while $F \to \|d\|$ near contact keeps (14) authoritative. Finally each target is clamped in front of the chassis front line (§2) so no arm can be commanded behind the torso.

Elbow-hint standoff (2026-10-16; applies to Step 8's output). Whatever produced the elbow hint e_r (the tracked forearm line or Eq. (17) below, mapped onto the robot's shoulder s_r), it is shifted outward before it enters the IK's elbow-direction cost:

$$e_r' = e_r + a\,\frac{\Pi_{xy}(s_r - s_r^{\text{other}})}{\|\Pi_{xy}(s_r - s_r^{\text{other}})\|}, \qquad a = \texttt{ARM_ELBOW_ABDUCT} = 0.15\,\text{m}, \tag{16}$$

with Π_{xy} the horizontal projection. Only the direction $e_r' - s_r$ is scored ($W_{\text{elbow}} = 15$ against the wrist's 600), so this is a soft abduction bias, strongest on a hanging arm and fading as the elbow rises; the wrist target is untouched. Reason: a human elbow hangs beside a torso about as wide as the shoulders, while the robot's shoulders are ∼ 3× narrower against its torso, so copying the operator's elbow direction parked the upper arm on the self-collision margin (operator on NOVA: "way too close to the body, it feels tucked"). Measured on three live Vision Pro takes through NOVA's deployed retargeter (E9 of the Retarget Lab doc, `web_gui/retarget_lab/experiments/exp_elbow_body.py`): median elbow clearance to the torso capsule 62–125 → 106–196 mm, upper arm within 20 mm of the torso on 3.6–11.2 → 0.0–0.7 % of frames, wrist error p50 27–38 → 23–33 mm. $a = 0$ restores the plain copy.

Step 8 — synthesised elbow (swivel from the palm roll). ARKit streams no tracked elbow (its forearm joint is a hand-anchor extrapolation), and with no elbow objective the redundant arm's configuration is weakly determined for high or close targets (measured 0.36–0.52 rad/tick basin flips). A two-link elbow is synthesised from the estimated shoulder, wrist, and HAND ORIENTATION, with $u = 0.56 L_h$ (upper arm), $f = 0.44 L_h$ (forearm), $\ell = \min(\|v\|, u{+}f)$. The law of cosines fixes the flexion; the swivel basis $(\hat{r}_0, \hat{t}_0)$ is gravity-referenced:

$$\cos\alpha = \frac{u^2 + \ell^2 - f^2}{2u\ell}, \qquad \hat{r}_0 = \frac{(I - \hat{n}\hat{n}^\top)(-\hat{u})}{\|\cdot\|}, \qquad \hat{t}_0 = \hat{n} \times \hat{r}_0,$$

$$e_h = s_h + u\cos\alpha\,\hat{n} + u\sin\alpha\,(\cos\varphi\,\hat{r}_0 + \sin\varphi\,\hat{t}_0), \tag{17}$$

so $\varphi = 0$ is the relaxed elbow-down pose. The swivel φ is estimated from the PALM'S ROLL about the arm axis (2026-07-22, "when I push my elbow up the sim does not follow"): raising the

elbow rotates the humerus and the whole distal arm rolls with it — and from a palm-down posture pronation is already near-saturated, so the palm *must* roll with a rising elbow. With $\hat{n}_p$ the palmar normal (the LEFT side's mirrored normal is flipped first) projected off the arm axis,

$$\theta = \operatorname{atan2}\big((\hat{r}_0 \times \hat{p})\cdot\hat{n},\ \hat{r}_0\cdot\hat{p}\big), \qquad \varphi_{\text{raw}} = \operatorname{clip}\big(\operatorname{sgn}\theta\ \max(0, |\theta| - \theta_{\text{dead}}),\ \pm 2.6\big), \tag{18}$$

where $\theta_{\text{dead}} = 60°$ deadbands the pronation/supination slack that rolls the palm with NO elbow motion (the formal under-determination of wrist-only data). φ is EMA-smoothed ($\alpha = 0.1$/frame; the raw per-frame roll wobbles with hand geometry and unfiltered it tripled the circle direction RMS), and the elbow objective adapts to it: $W^{\text{eff}}_{\text{elbow}} = W_{\text{elbow}}(1{+}1.5|\varphi|)$ and the Arm-3 shoulder-roll regulariser relaxes $\propto |\varphi|$ — at a large raise the rolled palm orientation is UNREACHABLE (J5/J6 saturate) and its cost exactly cancelled the elbow gain (two equal-cost basins, the arm stalling halfway); when the operator steers the elbow, its look wins the tie, while a neutral swivel keeps the original balance (fast-circle tracking unchanged at 3.8° RMS). Finally a TORSO-CLEARANCE clamp solves $\lambda(\varphi) = P\cos\varphi + Q\sin\varphi + R = \lambda_{\min}$ exactly (the root nearest the desired φ; an earlier stepped walk made the hint jump discretely) so the elbow never crosses inward past the shoulder — complementing the new `l/r_upper`↔torso pairs added to the self-collision guard (the upper arm was previously entirely unguarded). These are ELBOW SPHERES ($r = 0.035$ m) with a 3 mm *per-pair* margin, and the geometry forced both choices: the settled right elbow rests only 0.136 m from the torso axis, so an upper-arm segment capsule sits 6 cm *inside* the torso capsule at rest, and the global 2.5 cm margin spring-loads the natural rest pose (measured: 15° of fake "sag" in impedance holds as the torque-domain barrier pushed a merely-hanging arm), while a zero margin let the discrete CBF overshoot ∼1 mm negative under an adversarial crossing. The hint maps like every other vector, $e_{\text{rob}} = s_r + M_t(e_h - s_h)$, feeds only the direction cosine of (4), is overridden by any tracked (webcam) elbow, and is gated OFF when $\|d\| < 0.35$ (near bimanual contact it measurably fought closure by 3 cm). Validation (E7): the robot's elbow swivel follows a commanded 0 to −115° sweep with 23–40° error — the robot's INTRINSIC capability is itself ∼20° short at mid swivels (probed with a dominant W_{elbow}=600: J5/J6 saturate holding the rolled palm) — the elbow visibly rises from −16 cm below the shoulder to 4 cm above, and a cross-body reach holds the upper-arm capsules at the QP's exact 2.5 cm margin with zero solver fallbacks.

Step 9 — palm orientation. The operator's natural palm frame F_{op} (§3) is carried through the SAME map, $\hat{f}_{\text{tgt}}, \hat{n}_{\text{tgt}}$ from $M_t F_{\text{op}}$ — with the fixed world map a turned operator's palm axes stayed world-anchored and the hands twisted with the body heading.

Step 10 — the arm solve. The targets $(c, e_{\text{rob}}, \hat{f}_{\text{tgt}}, \hat{n}_{\text{tgt}})$ enter the cost (4), minimised by the projected damped Gauss–Newton of the *Solver performance* paragraph: stacking the weighted residual rows (using $1 - \hat{a}^\top\hat{b} = \frac{1}{2}\|\hat{a} - \hat{b}\|^2$ to express the cosine terms as least squares)

$$r(q) = \begin{bmatrix} \sqrt{W_{\text{wrist}}}\,\big(c^{\text{rob}}(q) - c\big) \\ \sqrt{W_{\text{elbow}}/2}\,\big(\hat{e}^{\text{rob}}(q) - \hat{e}\big) \\ \sqrt{W_{\text{orient}}/2}\,\big(\hat{f}^{\text{rob}}(q) - \hat{f}_{\text{tgt}}\big) \\ \sqrt{W_{\text{orient}}/2}\,\big(\hat{n}^{\text{rob}}(q) - \hat{n}_{\text{tgt}}\big) \\ \sqrt{W_{\text{smooth}}}\,\big(q - q_{\text{prev}}\big) \\ \sqrt{W_{\text{reg}}} \odot \big(q - q_{\text{nom}}\big) \end{bmatrix}, \qquad \big(J^\top J + \lambda I\big)\,\Delta q = -J^\top r, \tag{19}$$

with every Jacobian block analytic (`mj_jac`; unit-vector blocks via $J_{\hat{v}} = (I - \hat{v}\hat{v}^\top)J_v/\|v\|$, rotating-axis blocks via $\partial(Ra)/\partial q_j = J_{\omega,j} \times (Ra)$), $\lambda = 10^{-2}$, bounds handled by the active-set reduction,

and warm solves trust-region bounded to $q_{\text{prev}} \pm 0.10$ rad (a 6 rad/s slew — far above teleop speed, but inter-frame basin hops become impossible). The engagement-ease clock is separately clamped to 40 ms/tick so a wall-clock stall can never fast-forward the safety ease (a measured 1.5 s stall once published a 1.2 rad step).

Step 11 — basin escape (history independence). A long excursion can strand the warm-started redundant arm in a poorer local minimum which the trust region then faithfully preserves (measured 81 mm residual after a walk loop). Whenever the target has stayed inside a 1 cm ball for 12 frames, the incumbent q is compared against a converged cold solve $q_\circ$ from q_{nom} over the full joint range, by total cost with the smoothness term neutralised (else it always vetoes the challenger):

$$E_\circ(q_\circ) < 0.98\, E_\circ(q) \implies \text{latch } q^\star = q_\circ \text{ and walk } q \to q^\star \text{ in trust-bounded steps.} \tag{20}$$

The latch is essential — a single steered step is pulled straight back by the incumbent minimum on the next warm solve — and is dropped the moment the target moves. Since $q_\circ$ is a deterministic function of the (body-relative) target, every quasi-static pose becomes history-independent: the property the E6 walk-loop test asserts.

Solver performance (2026-07-22). Profiling on the Mac (canonical model `mabel_full.xml`) took the whole-body retarget frame time from 33 ms p50 — an effective ∼30 Hz ceiling — to 6.0 ms p50 / 10.5 ms p95 (167 Hz p50 / 96 Hz p95). Two fixes live here. First, the cosine arm IK's SLSQP — whose finite-difference gradients cost nine full-model `mj_kinematics` evaluations each — was replaced by a projected damped Gauss–Newton on the *same* cost (4) using MuJoCo's analytic Jacobians (`mj_jac`): 0.2–0.6 ms per side versus 6.2–7.3, with verified-identical optima on static targets (8 outer iterations with early stop, damping 10^{-2}; `MABEL_ARM_SOLVER=slsqp` keeps the scipy reference for parity checks). An *active-set* pass proved essential: coordinates that hit their box bound are fixed at the bound and the reduced normal equations re-solved, so the step slides along the active face — naive clipped GN zig-zags and leaves a 12.4° circle-direction tracking lag at *any* iteration count, versus 3.9° with the pass (1.3° SLSQP reference; the residual is transient lag on a deliberately fast test trajectory — static accuracy is identical). Second, the engine's `ThreadPoolExecutor` over the two arm sides became inline sequential solves: with sub-millisecond solves the pool's dispatch and GIL handoff measured 5–6 ms per frame — 10× the work it parallelised (`MABEL_ARM_POOL=1` restores it for A/B; the finger-solver savings are documented in `hand_retargeting.pdf`). Separately, the canonical MJCF's arm position-servo gains were raised k_p 100 → 500, k_v 20 → 45 (`left_arm.xml`/`right_arm.xml`, 7 actuators each): on a 0.5 Hz / 0.3 rad sine track with gravity compensation the joint error fell from 5.87° to 3.33° p50 (15.1° to 5.6° p95) and the phase lag from 188 to 86 ms, with $\leq 4\,\%$ overshoot on a 0.5 rad step; the large-step t_{90} floors at ∼170–200 ms, set by `forcerange` saturation (±40/27/7 N m) — physics, not gains — and the WBC `position_controller` inherits these gains from the model by design (`pd.py` reads `gainprm`). The bridge loop itself was verified healthy: the facade substep fast path (stale frame, memoized) costs 0.038 ms and `mj_step` 3.3 ms at the 4 ms timestep (1.2× real-time) — the old 33 ms retarget spike was what starved the physics accumulator.

Global (body-frame) mapping, separation floor, and basin escape (2026-07-22). Three additions make the map *globally* consistent, motivated by a live report: after walking around and returning to face forward with the same reach, the retargeting failed to reproduce the configuration, and clearly-separated human wrists sometimes overlapped on the robot. *(i) Body-frame decomposition.* The wrist-from-shoulder direction, palm orientation axes and synthesised

elbow were mapped through the *fixed* ARKit→robot world rotation M, so the map depended on the operator's world heading (and ignored the chassis heading). They are now decomposed in the OPERATOR body frame (right/up/forward from the body-frame estimator — wrist-consensus-grounded, non-accumulating) and reconstructed in the ROBOT'S chassis body frame: $M_t = R^{\text{rob}}_{\text{sem}}(t)\, B_{\text{op}}(t)^\top$, both sides rotating with their own body, so the same body-relative pose maps to the same body-relative robot config for *any* operator heading, any walk history, and any chassis yaw (`MABEL_ARM_BODY_FRAME=0` reverts). *(ii) Anti-collapse separation floor.* The robot's shoulders are 0.148 m apart vs a human's ∼0.45 m, so inward-angled directions cross the midline far sooner on the robot — the geometric root of the overlap (measured: 1.6 cm robot separation at 30 cm human). The inter-wrist distance is now floored at $\min(0.18\,\text{m}, d_{\text{human}})$, symmetric about the pair midpoint: deliberately *absolute*, so one-arm tasks are never touched (a γ-proportional floor measurably contaminated them), while it fades to the operator's own 1:1 separation near contact so touching hands still touch. *(iii) Basin escape.* A long excursion can strand the warm-started redundant arm in a poorer IK basin that the trust region then faithfully preserves (measured 81 mm residual after a walk loop). When the target has been quasi-static (inside a 1 cm ball) for 12 frames, the incumbent is compared against a converged cold solve from the nominal posture (smoothness term neutralised); a genuinely better canonical basin is *latched* and walked to in trust-bounded steps — one-shot steering is pulled straight back by the incumbent minimum. Quasi-static poses thereby become history-independent.

5.2 Experiments (`exp_retarget_fidelity.py`)

Headless, deterministic (synthetic clock, fixed seeds), driven through the shipped engine's per-frame path on synthetic operators with known ground truth; measurements are FK of the commanded actuator targets. Statures 1.55/1.75/1.95 m; the head-map baseline uses its *calibrated* reach (what the 1-pose calibration measures for the same operator). All 34 assertions across E1–E10 pass.

The E8 *scale-fidelity* battery is the direct test of the live requirement — that the robot track the human's head–wrist–wrist *scale*. Over a library of eight bimanual poses it reports the scaled-keypoint position error (the GMR $E_{\text{g-mpbpe}}$ analogue: each wrist relative to the mid-shoulder root, scaled by σ and mapped, versus the robot's realised wrist relative to its root) and the inter-wrist scale ratio $\|c^r - c^l\|_{\text{rob}} / (\sigma \|w^r - w^l\|_{\text{hum}})$:

scale fidelity (mean over the pose library)	GMR	extension	head
scaled keypoint error (g-mpbpe analogue)	26 mm	182 mm	92 mm
p95	60 mm	191 mm	120 mm
inter-wrist scale ratio (ideal 1.0)	0.965	0.541	0.902
inter-wrist direction error	0.7°	4.3°	0.1°

The per-shoulder extension map, exact on extension, is by far the *worst* on scale: anchoring on the robot's narrow shoulders halves the inter-wrist separation. The GMR map is within 3.5 % of the ideal scale and 26 mm on the triangle. The reach/tracking metrics (unchanged by the map choice — they share Steps 1–11) remain:

metric (forward reach unless noted)	extension	head (legacy)
ρ_r at full human extension (mean over statures)	0.962	0.920
elbow interior angle at full extension (hw max 161.4°)	161.9°	122.8°
circle tracking ρ RMS (at ρ=0.7)	0.031	0.065
circle tracking direction RMS (GN solver)	1.6°	25.7°
robot inter-wrist gap at 5 cm human gap	5.0 cm	—

The trust-region limit cycle (the "very unstable" report), found and fixed. A live report of the whole-body retargeting being "very unstable" resolved to a redundant-arm *limit cycle*, exposed by the E10 held-pose *sweep*: a single static hold pose can land in a clean IK basin by luck (an earlier one-pose check read 0.00 mrad), so E10 holds a grid of 36 bimanual reaches and reports the worst per-joint command standard deviation. **16 of 36 poses oscillated** at ± 0.10 rad — exactly the trust step. The mechanism: the warm-started arm solve ran the Gauss–Newton iterations *inside* the trust box $q_{\text{prev}} \pm 0.10$, which re-centres on q_{prev} every frame; when the basin's true local optimum lay outside the box, the "converged" q was a box *edge* — not a stationary point — and that edge flipped sides each frame, a stable 2-cycle (confirmed: escape-off and smoothness $\times 10$ both left it; *trust-off* removed it, converging to a clean fixed point). The fix (`arm.py solve()`) is to run the GN on the *full* joint range from the warm start — a local method, so it stays in the current basin (the anti-hop property the trust region protects is kept: the E6 walk-loop still reproduces the active arm to 6.5 mm) — and clip the *result* to the trust box. On a static pose the optimum is then a constant, so the arm walks to it monotonically and *stops*. After the fix:

E10 stability (GMR map, 36-pose held-reach sweep)	before	after
poses that oscillate (> 2 mrad command std)	16 / 36	**0 / 36**
worst per-joint command std over the sweep	$\sim$50 mrad	1.77 mrad
conversational-trajectory wrist tracking RMS	—	16.9 mm
max per-tick joint step on the slow sweep	—	0.095 rad

As a bonus the fix *improved* tracking (warm solves now reach true optima, not box edges): circle-tracking direction RMS 3.8° $\rightarrow$ 1.6°, and the walk-loop idle arm settled from 16 mm/31° to 2.3 mm/0.4°.

Elbow human-likeness (E9). The arm must "bend like a human's, not twist at weird angles." The elbow-*direction* residual is dominated by robot kinematics, not any IK weight (raising W_{elbow} from 30 to 120 moves mean error only 28° $\rightarrow$ 24°, p95 flat at 65° — and $W \geq 45$ re-opens the limit cycle above), so the human-likeness lever is the swivel deadband θ_{dead} that maps palm roll to elbow swivel (18). Tightening it 1.05 $\rightarrow$ 0.45 rad (a human forearm pronates only $\sim \pm 25°$ with the elbow still) cuts the swivel-follow error 28.7° $\rightarrow$ 15.7° (E9 deadband ablation, GMR map) — the elbow now tracks a commanded raise instead of hiding it — while the wrist-yaw "twist" DoF stays inside its $\pm 25.8°$ anatomical clamp. The E6 *global-consistency* battery asserts the new invariants: the identical body-relative pose at world yaws 0/90/180/270° produces identical arm commands (max difference 0.000 rad; the world-fixed map fails at 3.142 rad); after walking a square, spinning 360° and returning, the ACTIVE arm reproduces to 8.4 mm wrist / 0.4° elbow / 1.5° palm (the idle arm's cost-flat rest micro-pose is bounded loosely at 50 mm); and the separation transfer $d_{\text{rob}}(d_{\text{hum}})$ is monotone with a minimum 18 cm robot gap while the human is clearly apart (> 25 cm), closing to 5.0 cm at a 5 cm human gap. The robot's elbow *hardware* range stops at $-95.7°$ (161.4° interior): at full human extension the extension map drives it to within 0.5° of that mechanical maximum — the residual bend is the OpenArm's geometry, not the retargeting. All 12 edge-case assertions pass: hyper-extension saturation, wrist-at-shoulder degeneracy, hand-dropout freeze + reacquire, ARKit origin jump (no snap; the shoulder-anchored map is translation-invariant, pose recovered to 0.004 rad), crouch invariance, long-arm self-calibration, immediate profile apply, headless frames, `lock_j7`, and co-existence with the self-collision guard. The end-to-end real-dispatch chain (`exp_real_retarget_chain.py`) passes 5/5 consecutive runs with every published command step-bounded.

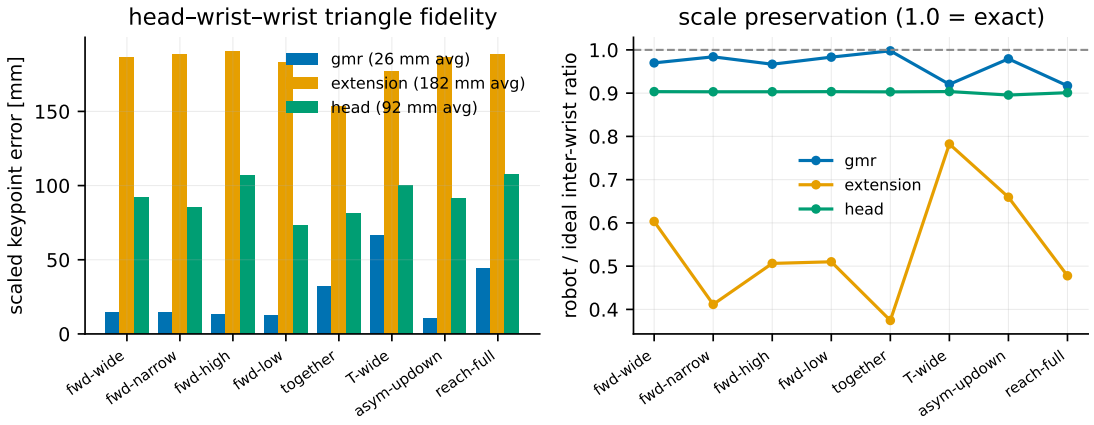


Figure 1: Scale fidelity (E8), the direct test of "track the human's scale." Left: per-pose scaled-keypoint error (head–wrist–wrist triangle) for the three maps over the pose library — the GMR map (blue) holds 26 mm average where the per-shoulder extension map (orange) is 7× worse. Right: inter-wrist scale ratio (1.0 = the operator's separation, scaled); the extension map collapses to 0.54 (the robot's narrow shoulders), the GMR map tracks 0.965.

6 The decoupled neck (`neck.py`)

Copying the operator's 6-DOF head pose onto a 3-DOF neck of different height is kinematically impossible; instead the head command reduces to a single 3D *look-at point* (the HoMMI relaxation: roll left free). The gaze ray (operator head rotation applied to the calibration gaze, made body-local so driving does not drag the gaze) is cast into the scene (`mj_ray`, self-hits within 25 cm rejected, 1.5 m fallback); the neck's yaw/pitch/roll targets then point the head's optical axis at the hit point by differential pointing with gain k_p=4. **Decoupling contract:** the controller reads only the head pose/scene and writes only the three neck actuators — asserted by `tests/test_retargeting.py` — so it composes with any arm/hand mode and steps aside entirely when a client drives the neck in joint space (the facade evicts stale neck joint targets when the head-follow owns it).

6.1 Live engine: gaze coverage and the neck-yaw→base-yaw handoff

The live engine (`teleop_engine/geometry.py::neck_angles`) maps the head's rotation relative to a latched, torso-relative datum R_0 directly onto the neck, after removing the *measured* chassis yaw ψ_c and torso pitch: $\Delta R = R_z(-\psi_c)\, M\, R_0^{\top} R_h\, M^{\top}$, decomposed as yaw–pitch–roll (M = ARKit→robot axes). The full-immersion Vision Pro view shows the head camera on a head-locked window, so the picture is *always* drawn where the operator looks. It is the right picture only while the neck can follow. Range (MJCF `ctrlrange`): yaw ±90°, pitch ±45°, roll ±45°.

Coverage, neck only (`experiments/exp_head_gaze_coverage.py`, part A). Synthetic ARKit head poses on a 5° grid (yaw ±180°, pitch −80°..+60°; 73 × 29 poses) are mapped through `neck_angles`, clipped exactly as the engine clips, and the `zed_left` optical axis is measured by MuJoCo FK in the chassis frame (torso and chassis at the home keyframe). Inside the range the camera follows the head *exactly* (0.0° error at pitch $-10/-20/-30/-45°$ and yaw 45/90°; a nod down moves the camera down 1:1). Outside it the error grows 1:1 with the excess: 15° at a 60° look-down, 30° at a 120° look left, 90° looking behind. Only 33.2 % of the grid (39.8 % of the

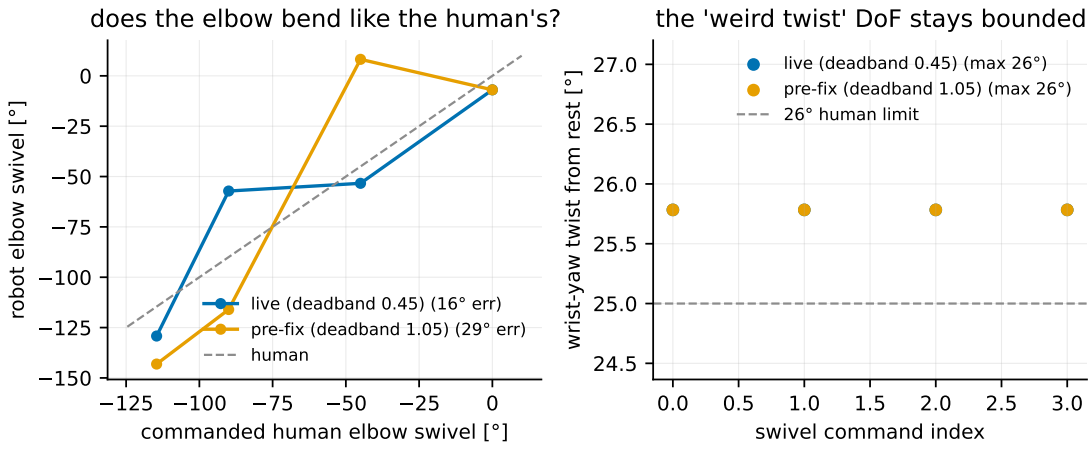


Figure 2: Elbow human-likeness (E9). Left: robot elbow swivel vs the operator's commanded swivel (identity dashed) — the tightened deadband (blue) follows a raised elbow that the pre-fix deadband (orange) hid. Right: the wrist-yaw "weird twist" DoF stays inside the ±26° human anatomical clamp for both.

sphere, area-weighted) is within 2° (Fig. 6).

The handoff (`teleop_engine/neck_handoff.py`, operator 2026-10-23: "I need to have 360 view as I move my head"). Let ψ_n be the *unclipped* neck yaw that `neck_angles` asks for, so $\psi_n = \psi_h - \psi_c$. Past a handoff angle $\psi^\star = 80°$ (10° inside the joint limit, so the neck still tracks a fast head while the base catches up) the excess becomes a base yaw rate:

$$e = \psi_n - \mathrm{clip}(\psi_n, -\psi^\star, \psi^\star), \qquad \omega_z = \mathrm{clip}(k\,e, -\omega_{\max}, \omega_{\max}), \tag{21}$$

with $k = 2\,\mathrm{s}^{-1}$ and $\omega_{\max} = 0.8\,\mathrm{rad/s}$, below `BASE_WMAX`. Because ψ_c is *measured* (encoders and odometry are mirrored into the model on the real robot, `server/plant.py`), the turn closes the loop by itself: as the chassis turns, $\psi_n \to \psi^\star$ and $e \to 0$. The rate is *added* at the point of consumption (`base_step(extra_wz=)`, never written back into the persistent `base_tgt`). It then passes through the tip-safe `BaseMotionModel` like every base twist. It is active in every scope, but never in DRIVE (the stick owns the base), while clutched, when the neck is frozen, or when not engaged and streaming. `MABEL_NECK_YAW_HANDOFF=0` disables it. *Assumptions:* the closed loop needs a live chassis-yaw measurement. *Stall guard:* the handoff integrates the yaw it commanded and the (unwrapped) yaw actually measured. Once commanded exceeds measured by 45° (stale odometry, a blocked base), it latches off until the next engage, clutch, or Set Forward. A rate law alone would otherwise spin the base indefinitely.

Closed loop in physics (part B). The operator snaps the head to ψ and holds it. Each 100 Hz tick runs `neck_angles` with the measured chassis yaw, the handoff, `BaseMotionModel`, `swerve_ik`, and `mj_step` on `scenes/mabel_bare.xml`:

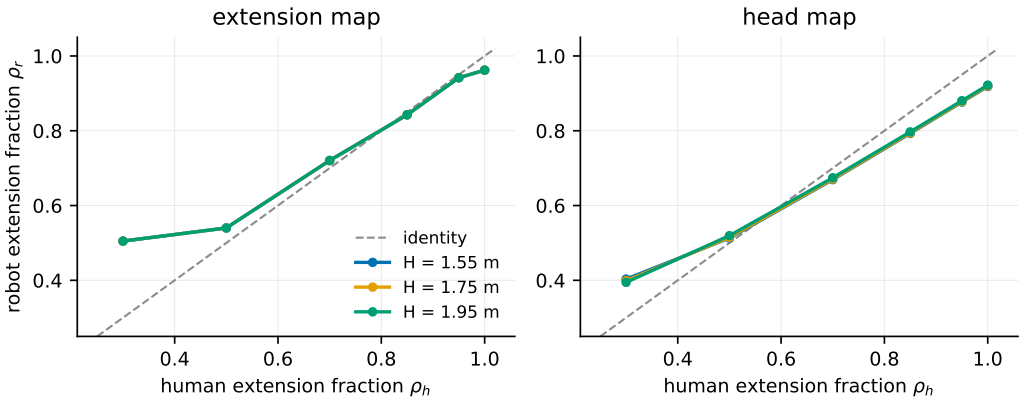


Figure 3: Extension transfer $\rho_r(\rho_h)$, forward reach, three statures. The extension map (left) is identity up to the hardware elbow stop for every operator size; the calibrated head map (right) distorts mid-workspace and saturates at $\rho_r \approx 0.92$ — the reported bent elbow. The GMR map inherits this full-extension guarantee via the renormalisation (6).

hold	t_{90} [s]	final error	neck yaw	chassis yaw	stalled
60°	0.23	0.2°	59.8°	0.0°	no
100°	0.28	0.2°	80.3°	19.9°	no
135°	0.76	0.2°	80.3°	54.9°	no
180°	1.65	0.2°	80.7°	99.6°	no

Every hold, including a look straight behind, converges to 0.2°, with the neck parked at the handoff angle and the chassis carrying the rest (Fig. 7). Unit contracts (no motion inside the range, sign, cap, closed-loop settle, a full 360° without a false stall, latch-off on stale odometry, reset re-arms) are in `tests/test_neck_yaw_handoff.py`. **Pitch has no handoff:** the base cannot pitch, so a look down past 45° stays capped (the error grows 1:1 past it, as in part A).

7 Lazy reach followers (`followers.py`)

MABEL's analogue of root-translation handling: the big, slow DOFs move only when the hand-target centroid c leaves a comfort box, each by a piecewise-linear deadband P-law

$$v \;=\; k\,(e - e_{\max})_+ \;-\; k\,(e_{\min} - e)_+, \tag{22}$$

applied to: the *base* (forward error over $[0, 0.80\,\mathrm{m}]$ from the chassis centre — the lower bound pinned to the body line so the base retreats the instant a wrist crosses behind the torso; lateral half-span 0.25 m; yaw deadband 0.30 rad); the *lift* (centroid height about a 0.63 m reference, half-span 0.12 m, rate capped at the firmware's 60 mm/s); and the *torso* (forward lean beyond 0.45 m of forward reach, before the base must translate). Follower base twists pass through `safe_base_step` downstream like every other twist.

7.1 Live engine torso law: box vs trunk angle

The live engine's cascade (`teleop_engine/wholebody.py`, `WholeBodyController.update_targets`) drives the torso from the operator's HEAD displacement (d, h) since calibration (forward, drop;

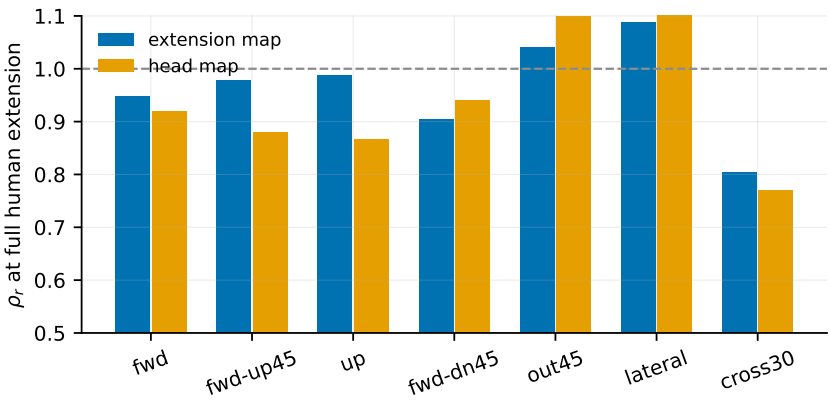


Figure 4: ρ_r at full human extension by direction. Lateral directions exceed 1 slightly (the true lateral reach exceeds the scalar L_r); cross-body is joint-limit bound for both maps.

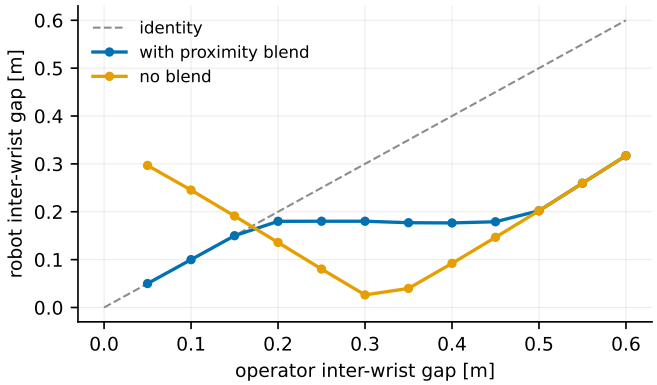


Figure 5: Bimanual closure: robot inter-wrist gap vs the operator's, with and without the proximity correction.

robot frame). The shipped *box* law is

$$q_{\text{torso}} = q_0 - k_t \ \text{db}\big(\,\text{clip}(d, -b_{\text{back}}, b_{\text{fwd}}),\ \delta\big), \qquad k_t = 2.5,\ b_{\text{fwd}} = 0.18,\ \delta = 0.06, \tag{23}$$

(db the soft deadband), which saturates at $k_t(b_{\text{fwd}} - \delta) = 0.30\,\text{rad}$ (17°) however deep the bend, on a joint that reaches 60°. On a live Vision Pro take the torso sat at exactly −0.300 while the operator's head went 0.23 m forward and 0.15 m down (`web_gui/retarget_lab/docs/retarget_lab.pdf`, E5). The *angle* law (`TORSO_LEAN_MODE=angle`) reads the trunk pitch from a hip-pivot model of length L (0.70 m, hip to eyes): a pitch θ moves the head $L\sin\theta$ forward and $L(1-\cos\theta)$ down, so

$$\theta_d = \text{atan2}(d,\ L - h), \quad \theta_h = \arccos(1 - h/L), \quad \theta = \max\big(\theta_d,\ \min(\theta_h,\ \arcsin(d/L_{\min}))\big), \tag{24}$$

and $q_{\text{torso}} = \text{clip}(q_0 - k_\theta\,\text{db}(\theta, \delta_\theta))$ with $k_\theta = 1$, $\delta_\theta = 0.08\,\text{rad}$. The forward reading is robust for small leans; the drop reading carries deep bends, in which the neck and lower spine also flex and the head falls further than a rigid hip pivot predicts (measured: $\theta_d = 22°$, $\theta_h = 38°$); the cap $\arcsin(d/L_{\min})$, $L_{\min} = 0.35\,\text{m}$, makes a squat (drop with no forward travel) a lift motion rather than a bend. Two couplings follow from the model: in GLOBAL scope the base walks only on the forward travel a

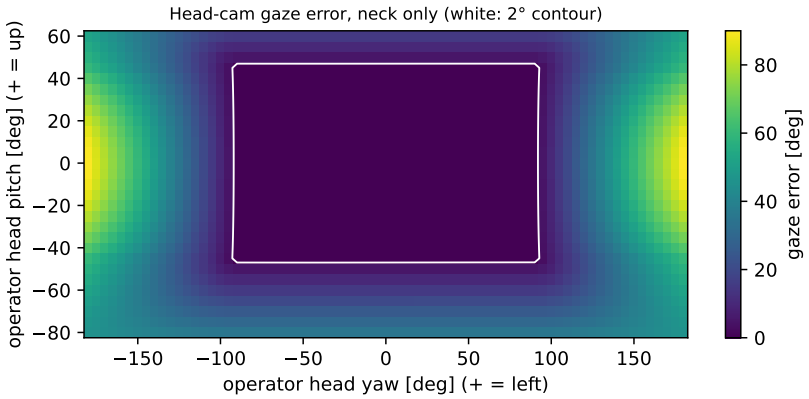


Figure 6: Head-camera gaze error over operator head yaw × pitch, neck only. The white contour bounds $\leq 2°$: the neck's $\pm 90° \times \pm 45°$ box.

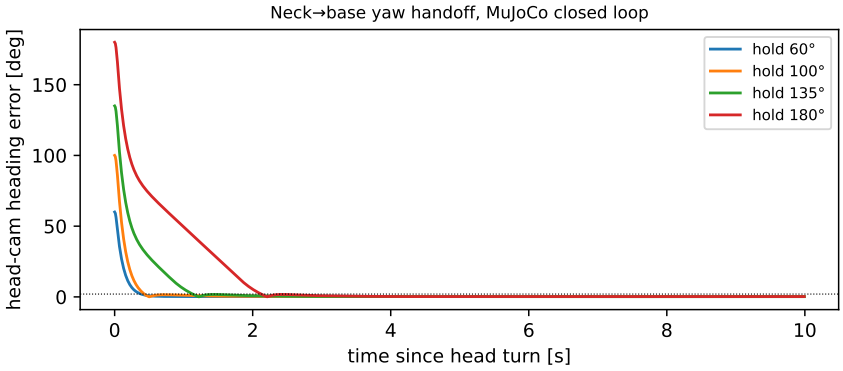


Figure 7: Neck→base yaw handoff in MuJoCo: heading error after the operator turns and holds the head. Dotted line: 2°.

bend does not explain, $d - \text{clip}(d, 0, L \sin\theta_h)$ (a walk keeps the head level, a bend lowers it); and the lift follows head height *plus* $L(1 - \cos\theta)$, since the robot's own torso bend reproduces that drop. Assumptions: the operator bends at the hips facing forward; a sideways lean or a crouch-and-reach violates the model (it degrades to the forward reading). Default `box` on the robot: a 60° real torso lean moves the CoM far forward and must be validated with `wbc_stability.py` first. The Retarget Lab runs `angle`. Pinned by `tests/test_torso_lean_model.py` (box saturation as the control, 20/40/55° bends, the measured deep bend, squat, walk vs bend in GLOBAL, lift).

8 Experiments

claim	source	result
mapping anchors and stays finite; $R \in SO(3)$; unit gaze	pytest	pass
neck writes only neck actuators	pytest	pass
followers silent inside the comfort box	pytest (facade e2e)	pass
end-to-end retarget under physics (V3 rig)	`exp_wbc_validation.py`	*currently failing*
per-operator scale recovery	`calibration.pdf`	Fig. 8

Honesty note: the V3 harness (`wbcval_retarget.pdf` traces) predates an in-flight engine change and fails at its priming assertion (`world ref never anchored`) *independently of* the 2026-07 package reorganisation (byte-identical before/after); it needs re-alignment with the engine before its tracking-error numbers are quotable again.

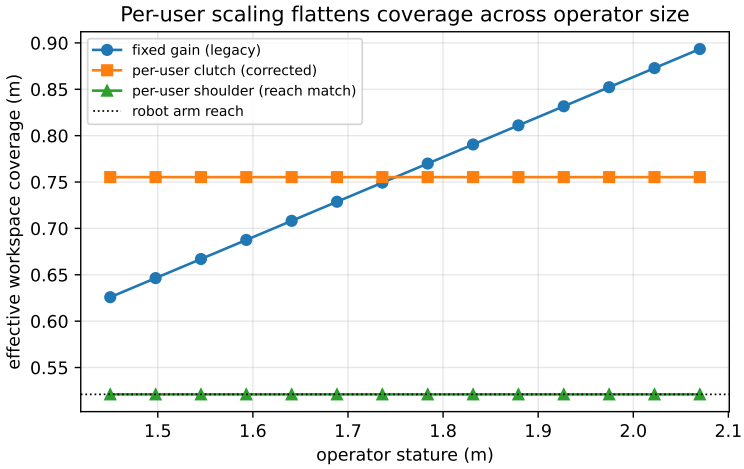


Figure 8: Per-operator workspace-scale recovery (from the calibration pipeline; derivations and full experiments in `calibration.pdf`).

8.1 Operator heading: the lazy-neck model

Both whole-body channels are referred to the operator's TRUNK heading: the base yaw follows it (GLOBAL scope) and the arm map's body frame is built on it (`ShoulderEstimator`, every scope). ARKit measures only the head. The live engine fused head and both wrists (`BodyFrameEstimator`); on real Vision Pro data (H1, below) it latched 30–75° offsets from where the operator faced after a hand-tracking dropout — it re-seeds from its own last state and has no absolute reference — and kept them for tens of seconds.

Model. The head yaw ψ_h is absolute and drift-free; the neck turns the head relative to the trunk within a comfortable range, and a head held turned means the trunk turned. With trunk estimate ψ_b and slack β,

$$e = \mathrm{wrap}(\psi_h - \psi_b), \qquad \psi_b \leftarrow \psi_b + \mathrm{sign}(e)\ \max(0, |e| - \beta), \tag{25}$$

$$e_{\mathrm{in}} = \mathrm{wrap}(\psi_h - \psi_b), \qquad \psi_b \leftarrow \psi_b + \left(1 - e^{-\Delta t/\tau}\right) g\, e_{\mathrm{in}}, \qquad g = \mathbb{1}\left[\,\overline{|\dot{\psi}_h|} < 20^\circ/\mathrm{s}\,\right], \tag{26}$$

with $\overline{|\dot{\psi}_h|}$ low-passed at 0.15 s. Properties (each a test in `tests/test_heading_and_follow.py`): (i) $|\psi_b - \psi_h| \le \beta$ at every instant; (ii) $\psi_b \to \psi_h$ exponentially (τ) whenever the head is held — no offset can persist, which is the defect; (iii) a turn larger than β is followed at once; (iv) a glance that returns within $\sim \tau$ moves ψ_b by at most $(1 - e^{-t_{\mathrm{held}}/\tau})$ of it, and not at all while the head moves; (v) hands are not used, so a hand dropout cannot corrupt it. Output in the engine's `body_yaw` convention (the robot-frame heading of the body's right axis, $\mathrm{atan2}(r_x, -r_y)$; π when facing ARKit $-z$). Assumption and its cost: the trunk eventually faces where the head is held. A seated operator who looks at a side task for several seconds turns the robot.

H1–H3 (`experiments/exp_heading.py`; three live Retarget Lab takes and 96 seated EgoDex clips, every 8th of the corpus). H1 *return*: the heading error wherever the operator faces their calibration direction again (head within 10°, held 3 s). H2 *follow*: $|\psi_b - \psi_h|$ during held turns ($> 60°$, 1 s). H3 *seated spread*: p95 of $|\psi_b - \psi_{b,0}|$ per clip, where the trunk is fixed by construction and the head looks around (head spread 2.9/12.1°, median/p90 clip). The head is the reference only where it is unambiguous: ARKit gives no trunk truth.

estimator	H1 return p90 [deg] live test1	H1 return p90 walking	H1 return p90 rotations	H3 seated spread p95 [deg] median / p90 clip (worst)
head+wrist (shipped)	13.3	72.7	51.0	3.1 / 14.4 (82.0)
lazy neck, $\beta = 25°$, $\tau = 2$ s	4.7	5.0	3.5	1.6 / 7.7 (23.9)

Table 1: H1/H3. The lazy-neck model against the shipped estimator.

Base follow, 1:1 in the body frame. With `BASE_TRANSLATION_MODE=body` the head displacement d (robot frame via `AVP_TO_ROBOT`) is rotated into the operator's current body frame, $d_b = R(-(\psi_b - \pi))\, d$; the comfort box and the lean are read on d_b; the overflow $o_b = d_b - \mathrm{clip}(d_b, \mathrm{box})$ is rotated out along the robot's heading, $R(\psi_{\mathrm{chassis},0} + \Delta\psi_b)\, o_b$, and added to the calibration base pose with gain 1. The shipped law (`shipped`) used a backward gain of 1.3, a reach multiplier up to 2× and the box in world axes: a lean after a turn read as a sidestep, and an operator not facing ARKit's $-z$ at calibration walked the robot partly sideways. The base yaw then tracks ψ_b with no deadband. Measured end to end in the Retarget Lab (its E6): heading p90 72→8° and position p90 0.69→0.34 m on the walking take.

Follow rates. The lift target was a per-frame low-pass (frame-rate dependent, ∼60 ms at 60 Hz, unbounded speed); it is now slewed at `LIFT_FOLLOW_SPEED` (0.03 m/s) and the torso lean at `TORSO_FOLLOW_SPEED` (0.5 rad/s, `config.MAX_TORSO_IK_SPEED`'s tip-over reason), whenever the engine passes its frame interval. Robot defaults: `HEADING_MODE=bodyframe`, `BASE_TRANSLATION_MODE=shipped` (unchanged); the rate limits apply everywhere.

9 Body-part ablation switches

For ablation studies the operator can freeze parts of the body *without changing the retargeter* (2026-10-16): Neck, Torso, and Upper body vs Whole body (upper = the lift *and* the mobile base frozen). The same engine runs in every combination; a frozen part is *parked* after it, by `ChannelPark` (`teleop_engine/safety.py`), which runs after the retargeter and the engagement supervisor and overwrites only its own actuators. On the wire these are the additive `control_mode.tracking` keys `neck`, `torso`, `lift` (plus `scope` for the base, §7).

Home. Neck: level and forward (all three axes 0). Torso: 0 rad. Lift: the home keyframe height (0.3175 m), so a parked lift is the same column in every trial rather than wherever it happened to be. Torso and lift parks act during retargeting only; in Drive the stick keeps the lift.

Freeze and release, both eased. With u_{live} the retargeter's command, a park started at t_0 from command u_a writes

$$u(t) = (1 - s)\, u_a + s\, u_{\mathrm{home}}, \qquad s = \min\big(1, (t - t_0)/T_p\big),\ T_p = 2.5\,\mathrm{s},$$

and a release started at t_1 from the last held command u_h writes

$$u(t) = (1 - w)\, u_h + w\, u_{\mathrm{live}}(t), \qquad w = \sigma\big(\min(1, (t - t_1)/T_r)\big),\ \sigma(x) = x^2(3 - 2x),\ T_r = 1\,\mathrm{s}.$$

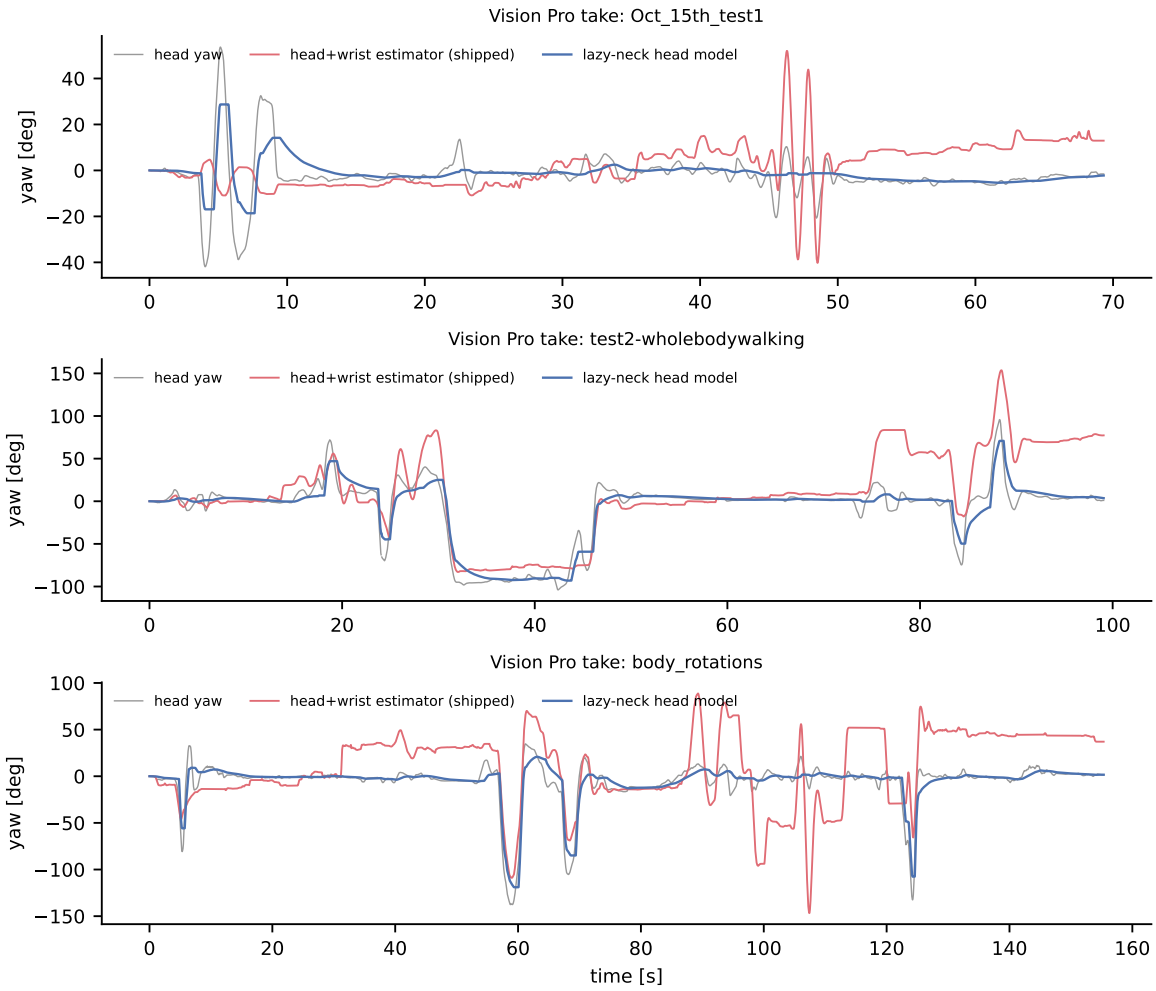


Figure 9: The three live takes: head yaw, the shipped head+wrist estimate (latched offsets after dropouts) and the lazy-neck estimate.

The release ease is new. Before it, re-enabling a channel simply stopped overriding, so the command stepped by the full live-minus-home difference in one tick (a neck re-enabled while the operator looked aside). A re-freeze mid-release anchors on the last command actually written; a unit test caught the first version anchoring on the old parked value and jumping (`tests/test_channel_park_release.py`, 4 tests).

Measured (`experiments/exp_ablation_toggles.py`, figure 11). The real facade (`MabelTeleop` → `TeleopEngine` → MuJoCo, `mabel_home.xml`) is driven by a synthetic operator who does everything at once: turns and nods the head (± 0.45 / ± 0.25 rad), leans, crouches and rises (± 0.10 m), steps forward past the comfort box and back (0–0.30 m), turns the body, and traces a bimanual circle. Each switch is applied at 2 s after an all-live warm-up, so every part has first been driven away from home; scored over 5.5–14 s on engaged frames.

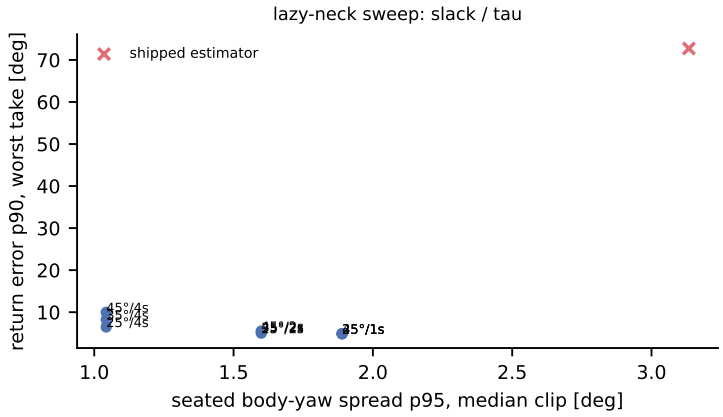


Figure 10: Sweep of slack $\beta \in \{25, 35, 45\}^\circ$ and $\tau \in \{1, 2, 4\}$ s: seated spread against the worst take's return error. The slack sets H2 exactly (the follow gap equals β while the head still moves); τ trades seated spread against return error. $25^\circ/2$ s chosen.

condition	neck range	torso range	lift range	base travel	arm range
whole body	71.9°	12.2°	104.6 mm	0.298 m	51.3°
freeze neck	0.22°	12.2°	104.6 mm	0.298 m	51.2°
freeze torso	71.9°	0.69°	104.9 mm	0.299 m	66.3°
upper body	71.9°	6.76°	1.2 mm	0.001 m	99.9°
all frozen	0.17°	0.50°	1.6 mm	0.003 m	100.6°

A frozen part's *command* sits exactly at home ($|u - u_{\text{home}}| = 0.000$ for neck, torso and lift); the realized ranges above (sub-degree, ≈1 mm) are the position servos' compliance under the arms' reaction torques. Unfrozen parts are unaffected by the others' switches (freeze neck leaves torso/lift/base at their whole-body values). *Same retargeter* is proven exactly, not statistically (`tests/test_ablation_same_retargeter.py`): each park writes only its own actuators, disjoint from every arm, wrist and finger actuator, and one engine tick from an identical state gives bit-identical arm commands with neck, torso and lift live or frozen. The end-to-end *arm range* column cannot carry that claim: it is a max-over-joints statistic of a wall-clocked loop and varies by about ±30° between *identical* runs (still body, 3 repeats each: whole body 56/109/59°, all frozen 48/31/93°; difference of means 17.5° against a pooled s.d. of 30.8°). Read it only as "the arms are retargeted in every condition". Physically, freezing the torso or the base under a leaning, stepping operator pushes the hand targets further from the robot's body, so larger arm excursions are expected there. *Bumpless*: peak commanded rate (per wall second) in the 3 s after a switch — freeze: neck 0.31, torso 0.13 rad/s; release: neck 0.31, torso 0.08 rad/s, lift 0.037 m/s (1 s average), i.e. the lift's own follow speed (0.03 m/s).

Harness conditions that changed the answer (each found by a wrong result): (i) the synthetic hands must carry metacarpals — the arm solver needs the palm frame (`geometry.operator_frame`) and *holds* the arm without one; bare wrists gave 0.0° of arm motion in every condition; (ii) a continuous walk pins the head at the comfort-box edge and saturates the torso bend, making "freeze torso" pass trivially; (iii) the loop runs well below real time while the supervisor and every follower clock on the wall — link-loss timeouts are scaled ×20 (in *every* namespace that bound them), only engaged frames are scored, and rates are per wall second (sim-time rates read ≈10× too fast); (iv) the lift follower steps once per operator frame, so its rate is averaged over 1 s.

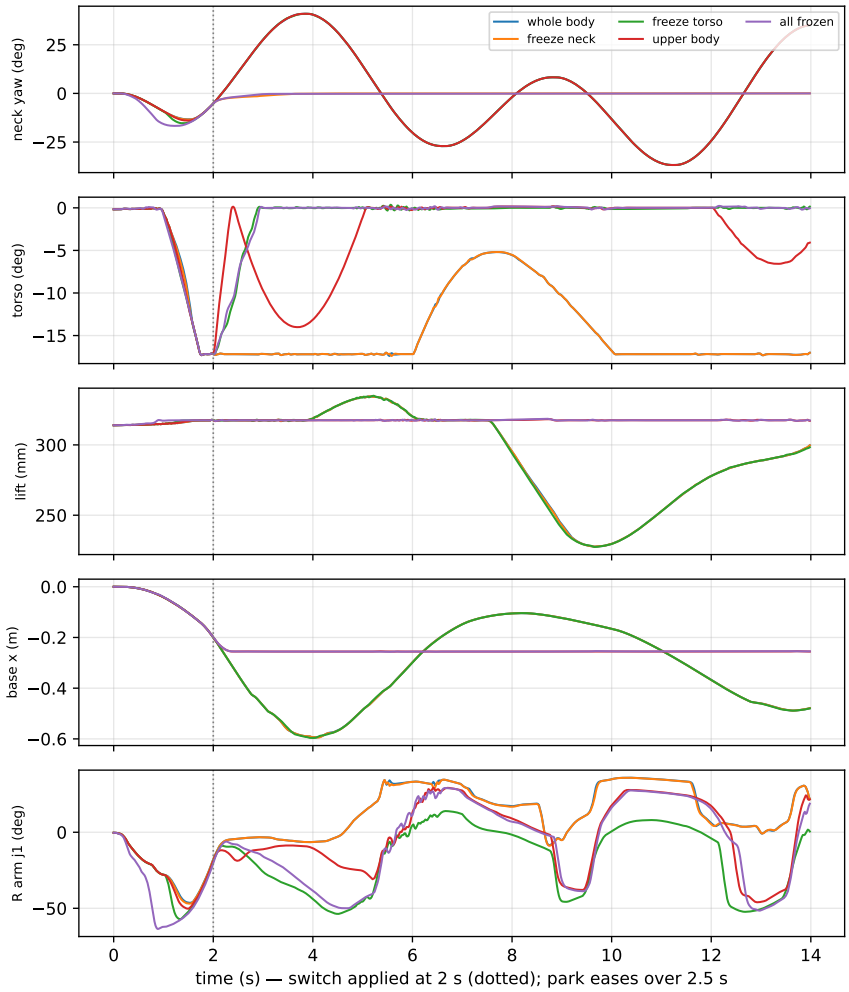


Figure 11: Neck yaw, torso, lift, base x and right-arm joint 1 under each switch (applied at 2 s, dotted). Frozen parts park at home over 2.5 s and hold.

Recorded. Every app-recorded episode stores the active switches per frame (`obs/control/*`, server `curation_recorder.py`).

Upper body tracks locally (2026-10-18). LOCAL scope used to read every forward head displacement as lean (d_{lean} = fwd), so an operator who walked 0.6 m forward bent the torso 0.63 rad and raised the lift 0.10 m. In LOCAL the lean is now read from the head drop h alone, past a gait-bob deadband $h_0 = 2$ cm: $\theta = \min(\arccos(1 - \max(h - h_0, 0)/L),\ \theta_{\text{cap}})$. The horizontal anchor is leashed to the forward distance that bend explains, $L \sin\theta$, so translation beyond it drags the anchor with the operator. The lift follows head height only in GLOBAL scope, and re-anchors on every scope switch. Synthetic operator (`web_gui/retarget_lab/experiments/exp_local_scope.py`, EgoDex pose, 1 s still, 3 s ramp, 1 s hold, 2 cm gait bob):

scope	operator	torso [rad]	lift [m]	base [m]	max arm $\lvert\Delta q\rvert$ [rad]
LOCAL	walk 0.6 m	0.000	0.000	0.000	0.021
LOCAL	hip lean 30°	−0.383	0.000	0.000	0.302
LOCAL	rise 0.15 m	0.000	0.000	0.000	0.036
GLOBAL	walk 0.6 m	−0.172	0.000	0.439	0.136
GLOBAL	hip lean 30°	−0.444	0.000	0.019	0.350
GLOBAL	rise 0.15 m	0.000	0.100	0.018	0.036

The drop deadband costs lean: a 30° hip bend gives −0.383 rad (22°) in LOCAL against −0.444 rad in GLOBAL. A bend that keeps the head height (pure forward reach without dropping) is not seen as lean in LOCAL.

Extended arms keep the operator's shape (2026-10-18). The GMR map $t = r_r + \sigma M(w - r_h)$ fixes the target's direction about the robot shoulder s_r by the root-uniform scaling, and with shoulders 3× narrower than a human's a straight arm is splayed outward. For an operator extension $\rho = |w - s_h|/L_h$ the target direction is now blended toward the operator's own, $\hat{u} = \text{normalize}\big((1-g)\,\widehat{t - s_r} + g\,M(\widehat{w - s_h})\big)$, with $g = \text{smoothstep}\big((\rho - 0.70)/(0.95 - 0.70)\big)$; the magnitude is unchanged. Below $\rho = 0.70$ the map is pure GMR, which keeps the hands' scaled separation (a gate from 0.55 dropped that ratio to 0.81 at $\rho = 0.75$). Synthetic straight arms (`web_gui/retarget_lab/experiments/exp_arm_shape.py`, both arms swung rigidly about the shoulder, 2 s ramp, 1 s hold, robot NOVA): robot arm direction error 1.5–8.8° (was 7.6–21°) forward, up, down and on diagonals; on the 2026-10-15 Vision Pro take (`exp_arm_shape_take.py`) 5.5° at $\rho \in [0.85, 0.95)$ and 6.8° above (was 14.6° and 14.4°), unchanged below 0.70, palm miss unchanged (7.0 cm mean). The elbow keeps ∼10° of bend at full extension (left up to 25°): on this arm a straight elbow needs joints 3 and 5 at their ±90° stops (the left elbow also at its +1.542 rad stop), and a larger reach overdrive made it worse (1.12: mean 15.7°), so it is a property of the geometry, not of the map.

Limitations. Simulation only; the synthetic operator is not a human (its head and hands are exact). Freezing the base is LOCAL scope: the joystick still drives it. A frozen lift does not follow a drive-pad lift command during retargeting (by design — the condition is "lift at home").

10 Implementation & usage

```
from mabel.whole_body_retargeting import Retargeter, NeckController
rt = Retargeter(model, data); rt.calibrate(pose)
targets = rt.update(pose)  # {l_pos, l_mat, r_pos, r_mat, gaze_dir, grasp_*}
neck.solve_and_apply(look_at_point(*neck.gaze_origin_dir(), model, data), dt)
```

The facade (`MabelTeleop`) composes all of this automatically; direct use is for experiments and new frontends. Legacy imports (`mabel.retarget`, `mabel.arm_retarget`, `mabel.head`, `mabel.lazy_base`, `mabel.calibration`) resolve here through shims.

11 Limitations

- The neck-yaw handoff (§6.1) is validated in MuJoCo and by unit tests only; on the real robot it depends on live odometry yaw reaching the mirror, and its stall guard is the only protection if that feed goes stale. Its *physical* verification on thor is pending. Looking down past the neck's 45° has no handoff and stays capped.
- Two arm-IK variants coexist (DLS vs cosine); behavioural parity is approximate and flagged per component in `mabel_controller.pdf`. The live map is the *GMR scaled-keypoint* map of §4

(`mabel/teleop_engine/engine.py`, `MABEL_ARM_MAP=gmr`); the extension (§5) and legacy head maps remain selectable. This package's `wrist_clutch.arm_hk` still implements the head-anchored map and is no longer what the live stack runs.

- The GMR map's single upper-body scale $\sigma = L_r/L_h$ assumes the robot's two arms are symmetric (they are, to 1 mm of reach); a strongly asymmetric embodiment would need per-arm σ and a per-body scale table as in the full GMR formulation [4]. The mid-shoulder root is head-derived, so it inherits the ShoulderEstimator's anthropometric priors until a measured profile is loaded (§5 Steps 2–3).
- A *shorter*-than-prior arm cannot read $\rho = 1$ without a measured profile (the anthropometric floor under-scales it); run the 3-pose body calibration for small operators.
- The trust region caps per-joint tracking at 6 rad/s; the overdrive makes lateral full extension read $\rho_r \approx 1.04$–1.09 (the scalar L_r under-states lateral reach); cross-body full extension is joint-limit bound at $\rho_r \approx 0.84$.
- The V3 physical validation harness is presently out of sync with the engine (above); until re-aligned, end-to-end tracking-error claims rest on the pre-drift measurements in `mabel_controller.pdf`.
- The look-at ray assumes a meaningful scene intersection; in a cluttered scene near-field geometry can capture the gaze (mitigated by the 25 cm self-hit rejection).
- Followers are P-laws with hand-tuned boxes; no formal stability proof of the follower/IK cascade is attempted (empirically stable across V-tests).

References

[1] C. Chi et al., "Universal manipulation interface," *RSS*, 2024 (relative/clutch mapping).

[2] S. Yang et al., "ACE: A cross-platform visual-exoskeleton for low-cost dexterous teleoperation," *CoRL*, 2024 (arXiv:2408.11805; workspace scaling).

[3] HoMMI: whole-body mobile manipulation from human demonstrations (look-at relaxation; CoM cost study in `wbc_stability.pdf`).

[4] J. P. Araujo, Y. Ze, P. Xu, J. Wu, and C. K. Liu, "Retargeting Matters: General Motion Retargeting for Humanoid Motion Tracking," *ICRA*, 2026 (arXiv:2510.02252). Root-uniform per-body keypoint scaling + two-stage differential IK; the source of MABEL's live scaled-keypoint arm map (§4).